

Software Requirements Specification (SRS)

SmartAssetX – Intelligent Office Asset Management System

1. Introduction

SmartAssetX is a web application that centralizes office asset management, employee requests, warranty tracking, asset transfers, notifications, and reporting into a single platform for organizations.

2. Problem Statement

Many organizations manage assets manually, resulting in poor tracking, delayed approvals, unmonitored warranties, and inefficient resource utilization. SmartAssetX provides a digital solution to manage the complete asset lifecycle.

3. Objectives

- Manage office assets efficiently.
- Assign assets to employees.
- Handle request and approval workflows.
- Track warranties and AMC contracts.
- Generate reports and notifications.
- Improve asset utilization and reduce costs.

4. Users

Admin: Manages users, departments, assets, warranties, and reports.

Manager: Approves requests, monitors department assets, and handles transfers.

Employee: Views assigned assets, requests equipment, reports issues, and returns assets.

5. Functional Requirements

User Login, Asset Management Module, Employee Dashboard, Request & Approval Workflow, Warranty & AMC Tracking, Asset Transfer Module, Notifications, and Reports.

6. Non-Functional Requirements

Secure authentication, responsive user interface, reliable PostgreSQL storage, scalability, maintainability, high availability, and efficient performance.

7. Database Tables

Users, Roles, Departments, Assets, Asset_Categories, Asset_Assignments, Asset_Requests, Asset_Transfers, Warranties, AMC_Contracts, Vendors, Notifications.

8. Technology Stack

Frontend: React.js, React Router

Backend: Java, Spring Boot

Database: PostgreSQL

Security: Spring Security + JWT

Tools: Maven, Postman, Git & GitHub

IDE: IntelliJ IDEA & VS Code

9. Unique Features

- Asset request and approval workflow.
- Warranty and AMC renewal reminders.
- Department-wise asset transfers.
- Real-time notifications.
- Asset utilization and cost analysis reports.
- Centralized office asset tracking.

10. Future Enhancements

Email notifications, mobile application support, RFID/barcode integration, AI-based asset demand prediction, multi-branch management, and cloud deployment.